

AFTER THE COLLAPSE

Northern cod · NAFO 2J3KL

can any catch policy hold the LRP?
the 1992 moratorium

$r = 0.2369 \cdot K = 5000$ kt (at bound)
the fitted surplus map

884.6 kt (2016) — a floor on SSB
limit reference point

kernels · boundaries · replays · selection
protocol frozen before any score

WHAT CATCH CAN HOLD

under the worst-case floor

the 10th-percentile class
worst-case productivity floor

largest robust constant catch

91.6 kt



0 kt: survival 0.91 · CI [0, 87.1]

120 kt: survival 0.65

zone rule holds the LRP itself
hold the safe set; the critical-
zero catch and the moratorium

WHEN CATCH CANNOT HELP

the map is expansive at the LRP

under the declared partial order
no non-BAU policy dominates BAU

at the 5th-percentile floor, $T = \infty$
positive-catch rules: empty

beyond 7 yr · $K \geq 2K^* = 1769.2$ kt
certified kernels: empty

than the frozen convention implied
the margin they must supply is smaller
the LRP is protected by good years

Robust Viability of the 2J3KL Limit Reference Point under a Surplus-Production Map: Policy Scoring, Expansion, and When Catch Cannot Help